

Cooperative On-Ramp Merging Control Model for Mixed Traffic on Multi-Lane Freeways

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Abstract—This paper proposes a hierarchical model for cooperative on-ramp merging control (CORMC) in mixed traffic with both connected automated vehicles (CAVs) and connected human-driven vehicles (CHVs). The upper-layer of the CORMC model employs an anticipatory position searching (APS) algorithm to determine the anticipatory positions at which merging vehicles (MVs) should merge from the on-ramp lane to the adjacent mainline lane, and to assign cooperative vehicles (CVs) for each MV. A collaborative utility choice (CUC) model is presented to determine the optimal maneuver of CVs to create proper gaps for MVs. The driver compliance rate is introduced to account for CHVs' unwillingness to follow the instructions given by the CUC model. The lower-layer comprises a cooperative merging control (CMC) model that ensures safe and smooth merging execution for MVs. Longitudinal and lane changing models are developed for mainline vehicles to facilitate an efficient and safe merging process. Simulation results show that the performance benefits of the CUC model are marginal when the CHV compliance rate is relatively low. However, the performance improvement is significant at higher compliance rates (>50%). Furthermore, the CORMC model has the potential to increase merging and mainline throughput by over 75% at sufficiently high CAV penetration rates. Comparison of three control strategies shows that the APS algorithm plays an important role in the CORMC model. A comparison with the Simulation of Urban Mobility (SUMO) indicates that the CORMC model significantly mitigates the propagation of congestion waves across varying levels of CAV penetration and on-ramp flow rates.

Index Terms—Connected automated vehicles (CAVs), cooperative on-ramp merging, mixed traffic flow, multi-lane traffic, hierarchical framework.

I. INTRODUCTION

FREEWAY on-ramp merging areas are widely recognized as major bottlenecks that significantly impact the safety, efficiency and reliability of freeway traffic operations. At the

upstream of the freeway merges, the slow moving vehicles caused by merging vehicles (MVs) can propagate upstream to the mainline, resulting in congestion and an increased risk of crashes and conflicts [1], [2]. However, with the development of communication, sensing and control technologies, connected automated vehicles (CAVs) have the potential to alleviate congestion at freeway merging areas [3], reduce energy consumption and emissions [4], and enhance merging efficiency and safety [5]. Nonetheless, in the foreseeable future, freeway traffic will inevitably experience a mixed traffic state where both traditional human-driven vehicles (HVs) and CAVs coexist. Thus, it is of great significance to investigate efficient merging strategies for the mixed traffic flow on freeways.

In recent years, considerable research has been devoted to on-ramp merging control algorithms for pure CAV traffic with objectives such as minimizing fuel consumption, travel time, passenger discomfort [6], [7], [8], [9], [10], or improving traffic safety and efficiency [11], [12]. For instance, Ntousakis et al. [7] present a longitudinal trajectory planning method to assist the merging of vehicles on freeways, using a scenario of 5 vehicles with a single main lane to demonstrate the validity of their method. Scholte et al. [13] propose a cooperative strategy for a single AV merging into a vehicle platoon. However, their proposed method does not explicitly consider the merging sequence of multiple vehicles and therefore cannot adequately address the situation of multiple AVs merging into the platoon. Unlike the above studies, which mainly focus on merging trajectory planning and control, Jing et al. [9] propose a cooperative multi-player game-based optimization framework to globally optimize the merging sequence and trajectory for each vehicle. The simulation results demonstrate that their proposed model can reduce fuel consumption and travel time. Later, Chen et al. [11] develop a hierarchical control approach for efficient and safe merging operations of CAVs, where a tactical layer controller determines the optimal vehicle merging sequence, while an operational layer controller is designed based on model predictive control (MPC) to optimize the desired acceleration of CAVs.

Previous studies have developed approaches and algorithms for the cases with a single main lane, which do not consider lane-changing decisions and maneuvers for the mainline traffic. However, when on-ramp CAVs merge into a multilane freeway section, vehicles in the main lanes have the option to keep longitudinal movement or change lanes. Algorithms that only allow vehicles to keep longitudinal

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movement in their original lane cannot fully utilize the capacity of the other lanes, thereby reducing merging efficiency. To address this limitation, researchers propose cooperative merging control algorithms for multilane merging areas to increase throughput, reduce fuel consumptions, delays or travel times [14], [15], [16], [17], [18], [19].

Notably, most existing studies focus on pure CAV traffic where all vehicles are fully controllable. Limited research has investigated merging control in mixed traffic with CAVs and HVs. Wei et al. [20] develop a merging decision model that considers the interaction between the merging vehicle and the following vehicle in the main lane. Though the heterogeneity of AVs in terms of different driving styles is considered, the heterogeneity of HVs and their interactions are not explicitly taken into account. Pueboobpaphan et al. [21] consider mixed traffic with manual vehicles and cooperative adaptive cruise control (CACC) vehicles in the mainline and propose a merging assistant algorithm to control the mainline vehicles to create proper gaps for the on-ramp manual vehicles. However, their strategy cannot align the on-ramp manual vehicle with the assigned gap, resulting in only marginal improvement of efficiency (e.g., vehicle km-traveled and average travel time) for a given percentage of CACC vehicles. Karimi et al. [22] propose cooperative merging control algorithms for triplets of vehicles under different combinations of CAVs and HVs, with an optimal trajectory determined for each vehicle in the triplet considering safety and comfort. Similarly, Sun et al. [23] develop a bi-level cooperative control model that considers interactions between HVs and CAVs for a triplet of vehicles in a single main lane and one on-ramp lane, and validate their model's effectiveness with microscopic simulation. Mu et al. [24] present an event-triggered rolling horizon-based trajectory planning approach for the merging process of two platoons, aiming to improve traffic efficiency and smoothness, instead of independently addresses each individual merging maneuver.

Most state-of-the-art studies focus on pure CAV environment or mixed traffic situations where the merging area comprises only a single main lane and an on-ramp lane. Consequently, lane-changing decisions and maneuvers are not considered for mainline traffic, and the heterogeneity of HVs and CAVs in the mixed traffic flow is not fully addressed. Although some studies [14], [15], [16], [17], [18], [19] have dealt with the merging scenarios involving multilane situations, they primarily focus on optimizing the longitudinal trajectories of mainline vehicles using global optimization methods that require centralized controllers or road units to regulate the speed of each vehicle, leading to high computational and communication costs. Moreover, while lane-changing advice strategies for mainline vehicles are used to facilitate the merging process in [14], [15], [16], [17], and [18], the lane-changing maneuvers are determined without considering their impact on the vehicles upstream of the target lane, potentially affecting the safety of the vehicles upstream of the target lane.

To address the aforementioned issues, this paper proposes a hierarchical distributed framework for the cooperative on-ramp merging control (CORMC) model for mixed traffic

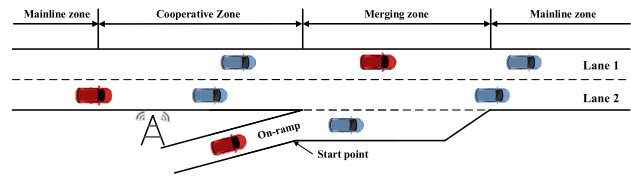


Fig. 1. The scenario of a freeway section with two main lanes in one direction and an on-ramp.

on multi-lane freeway merging sections, based on our previous works [25], [26], [27]. The main contributions of this study are as follows: (1) The proposed hierarchical distributed control approach ensures efficient and safe merging of vehicles in the multi-lane merging zone under mixed traffic environments, explicitly considering the lane changing maneuvers in the main lanes. (2) Two collision-avoidance algorithms, the boundary-collision-avoidance and front-collision-avoidance algorithms, are developed to ensure collision-free merging and lane changing processes. (3) A vehicle generation model is designed to describe the stochastic properties of the mixed traffic flow, taking into account the heterogeneity of HVs in terms of the desired speed and the heterogeneity of CAVs in terms of inertial lag. (4) The driver compliance to the suggestions provided by the proposed model is explicitly considered, and thus the impact of driver behavior on the performance of the proposed control approach can be effectively evaluated.

The remainder of the paper is organized as follows. Section II presents the proposed CORMC model framework, which is formulated as a hierarchical control framework comprising the upper decision-making layer and the lower control layer. Sections III and IV present the upper and lower layers of the CORMC model, respectively. In section V, a vehicle generation model is presented, and scenario settings and model parameters are given. In section VI, the simulation results are analyzed and discussed. Section VII concludes the paper, and provides some prospects for future work.

II. COOPERATIVE ON-RAMP MERGING CONTROL (CORMC) MODEL FRAMEWORK

This study considers a freeway section consisting of two main lanes in one direction and an on-ramp lane, as shown in Fig. 1. The section is divided into three areas: the main-line, cooperative and merging zones. The red vehicles represent CAVs, while the blue vehicles represent HVs. In the cooperative zone, the vehicle in the on-ramp lane searches for an anticipated merging position in lane 2. The on-ramp vehicle then cooperates with the vehicles in front or behind the anticipated merging position to reserve a safe merging spacing. In the merging zone, the on-ramp vehicle performs the merging maneuver until it reaches the center of lane 2.

Before presenting the framework of the entire CORMC model, some assumptions are made as follows:

- 1) All vehicles on all lanes are connected vehicles, and can communicate with each other and receive real-time state information via Vehicle-to-Vehicle (V2V) communication without considering communication delays. Throughout this paper, all HVs refer to Connected Human-Driven Vehicles (CHVs).

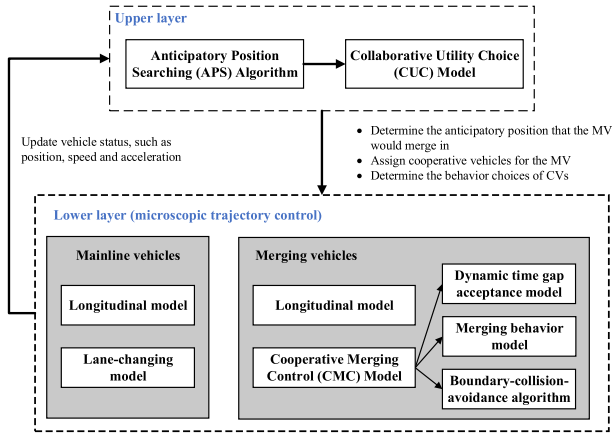


Fig. 2. Hierarchical framework of the CORMC model.

- 2) The behavior of CAVs is assumed to be fully controllable and deterministic, whereas CHVs are uncontrollable. In this study, CHVs are classified into two categories: compliant and non-compliant CHVs. The former adopts favorable suggestions given by the proposed model, while the latter does not. Thus, the mixed traffic flow consists of CAVs, compliant CHVs and non-compliant CHVs. It is noteworthy that non-compliant CHVs can be regarded as HVs lacking V2V communication.

Fig. 2 illustrates the proposed hierarchical framework of the CORMC model. The framework is distributed, meaning that each vehicle is controlled by its own controller, and multi-vehicle cooperative control is achieved through the information exchange between the vehicles. The upper layer of the framework consists of two parts: the anticipatory position searching (APS) algorithm and the collaborative utility choice (CUC) model. The APS algorithm determines the anticipatory positions at which the merging vehicles (MVs) merge from the on-ramp lane to lane 2 (as shown in fig. 1), and also determines the cooperative vehicles (CVs) in lane 2 for each MV. The CUC model allows the CVs in lane 2 determined by the APS algorithm to make the best choice between maintaining longitudinal movement and changing lanes. At the lower layer, the vehicle status, including position, speed, and acceleration, is updated for each time step using the longitudinal and lane-changing models based on the output decisions from the upper layer. The updated vehicle status information is then feedback to the upper layer to make decisions for the next time step. For the mainline vehicles, the microscopic trajectories are updated using the longitudinal and lane-changing models. For the MVs, a cooperative merging control (CMC) model is developed to coordinate the movements of vehicles in the on-ramp with vehicles in the adjacent lane to achieve a safe and efficient merging process.

III. UPPER LAYER OF THE CORMC MODEL

A. Anticipatory Position Searching (APS) Algorithm

In order to ensure safe and efficient merging of MVs onto the main road (lane 2), it is crucial to determine the

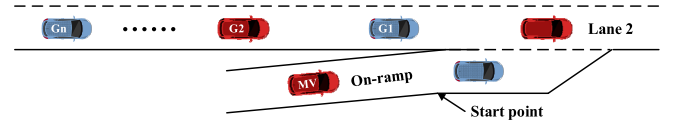


Fig. 3. Schematic illustration of the merging area including the on-ramp and lane 2.

anticipatory position of the MV in lane 2 in advance. The two vehicles in front and behind the anticipated merging position, called cooperative vehicles (CVs), then cooperate to reserve the safe merging spacing for the MV. To achieve this, we have developed an APS algorithm which judges the MV's suitable position and assigns CVs for each MV.

Fig. 3 illustrates the merging zone, including the on-ramp and lane 2. We define the set of vehicles which may create a merging gap for the MV at time t as:

$$Gap_car_{MV}(t) = [G_1, G_2, \dots, G_{n(t)}] \quad (1)$$

with constraints:

$$\begin{cases} x_j(t) \in [x_{MV}(t) - L^{cr}, x_{MV}(t) + L^{cr}], j \in Gap_car_{MV}(t) \\ x_{G_1}(t) > x_{G_2}(t) > \dots > x_{G_{n(t)}}(t) \end{cases} \quad (2)$$

where G_j represents vehicle j within vehicle set $Gap_car_{MV}(t)$. $n(t)$ represents the total number of vehicles that may produce a merging gap for the MV at time t . $x_j(t)$ and $x_{MV}(t)$ represent the longitudinal positions of vehicles j and the MV at time t , respectively. L^{cr} represents the length of the communication range, and all vehicles in Gap_car_{MV} are within the communication range L^{cr} of the MV. In this study, L^{cr} is set to 300m.

The core idea of the APS algorithm is that the MV calculates the anticipatory arrival positions of all Gap_car_{MV} vehicles traveling at their current speed for time $T_{MV}^*(t)$. $T_{MV}^*(t)$ represents the anticipatory time required for the MV to travel at a constant speed of $v_{MV}(t)$ from the current position to the start point of the merging zone at time t . The anticipatory longitudinal spacing $\mathcal{D}_j^*(t)$ between the anticipatory arrival position of vehicle j and the start point of the merging zone at time t is then calculated as:

$$\mathcal{D}_j^*(t) = x_j^*(t) - x_0^m - L_j \quad (3)$$

$$x_j^*(t) = v_j(t) \cdot T_{MV}^*(t) + x_j(t), j \in Gap_car_{MV}(t) \quad (4)$$

$$T_{MV}^*(t) = \frac{x_0^m - x_{MV}(t)}{v_{MV}(t)} \quad (5)$$

$$\mathcal{D}_{MV}(t) = [\mathcal{D}_1^*(t), \mathcal{D}_2^*(t), \dots, \mathcal{D}_j^*(t), \dots, \mathcal{D}_{n(t)}^*(t)] \quad (6)$$

where x_0^m represents the longitudinal coordinate of the start point of the merging zone. $x_j^*(t)$ represents the anticipatory position of vehicle j after traveling at a constant speed of $v_j(t)$ for a time period of $T_{MV}^*(t)$. $\mathcal{D}_j^*(t) > 0$ indicates that vehicle j travels past the start point of the merging zone after time $T_{MV}^*(t)$ while the MV is at the start point of the merging zone. L_j is the length of vehicle j . $\mathcal{D}_{MV}(t)$ is the set of anticipatory longitudinal spacings of all vehicles in $Gap_car_{MV}(t)$ at time t .

If the conditions that $\mathcal{D}_j^*(t) > 0$ and $\mathcal{D}_{j+1}^*(t) < 0$ are satisfied, it indicates that the vehicles preceding vehicle j have

passed the starting point of the merging zone by the time the MV reaches it, while the vehicles behind vehicle j are all located upstream of the starting point of the merging zone. Then, vehicle j is designated as the anticipatory cooperative leading vehicle (CLV), and the vehicle that follows vehicle j is designated as the anticipatory cooperative following vehicle (CFV). Correspondingly, the position between the CLV and CFV is considered as the anticipatory position of the MV in lane 2.

Let $\mathcal{D}_{CLV}^*(t)$ and $\mathcal{D}_{CFV}^*(t)$ denote the anticipatory longitudinal spacings of the CLV and CFV at time t , respectively. Let $\mathcal{D}_{min}^{CLV}(t)$ and $\mathcal{D}_{min}^{CFV}(t)$ denote the minimum acceptable spacing between the MV and CLV, the minimum acceptable spacing between the MV and CFV at time t , respectively. The constraints for the anticipatory longitudinal spacings of the CLV and CFV are given by:

$$\begin{cases} \mathcal{D}_{CLV}^*(t) \geq \mathcal{D}_{min}^{CLV}(t) \\ |\mathcal{D}_{CFV}^*(t)| \geq \mathcal{D}_{min}^{CFV}(t) \end{cases} \quad (7)$$

$$\begin{cases} \mathcal{D}_{min}^{CLV}(t) = v_{MV}(t) \cdot g_{min}^{CM} \\ \mathcal{D}_{min}^{CFV}(t) = v_{CFV}(t) \cdot g_{min}^{CM} \end{cases} \quad (8)$$

where $v_{MV}(t)$ and $v_{CFV}(t)$ represent the speeds of the MV and CFV at time t , respectively. g_{min}^{CM} represents the anticipatory minimum acceptable time gap of cooperative merging.

The collaborative attribute of CVs is given by:

$$col = \begin{cases} 0, & \text{no collaboration is required} \\ 1, & \text{collaboration is required} \end{cases} \quad (9)$$

where 0 and 1 indicate that the coordinated maneuver is not required and coordinated maneuver is required, respectively. According to the constraints in Eq. (7), we can divide the cooperative situation into four cases.

Case 1: Both constraints in Eq. (7) are satisfied.

In this case, when the MV travels to the start point of the merging zone, the distance between the CFV and CLV ensures that the MV can merge into lane 2 smoothly and safely if all three vehicles drive at the current speed. Therefore, no collaborative maneuvers are needed and the collaborative attributes $col_{CLV} = 0$ and $col_{CFV} = 0$.

Case 2: Only the first constraint in Eq. (7) is satisfied.

In this case, the anticipatory longitudinal spacing between the MV (at the start point of the merging zone) and CFV is smaller than the minimum acceptable spacing. Therefore, the CFV needs to perform collaborative maneuvers based on the CUC model, which will be introduced in detail in section III-B. The collaborative attributes $col_{CLV} = 0$ and $col_{CFV} = 1$.

Case 3: Only the second constraint in Eq. (7) is satisfied.

In this case, the anticipatory longitudinal spacing between the CLV and MV (at the start point of the merging zone) is smaller than the minimum acceptable spacing. Therefore, the CLV needs to perform collaborative maneuvers based on the CUC model. The collaborative attributes $col_{CLV} = 1$ and $col_{CFV} = 0$.

Case 4: Both constraints in Eq. (7) are unsatisfied.

In this case, both the CLV and CFV need to perform collaborative maneuvers based on the CUC model. The collaborative attributes $col_{CLV} = 1$ and $col_{CFV} = 1$.

In cases 2 and 4, to ensure that the CFV can adjust the required merging gap for the MV to satisfy both constraints in Eq. (7), the desired car-following spacing of the CFV is calculated according to:

$$S_{CFV} = \mathcal{D}_{min}^{CFV}(t) + L_{MV} + \max(\mathcal{D}_{min}^{CLV}(t), \mathcal{D}_{CLV}^*(t)) \quad (10)$$

In case 3, although the spacing between the MV and CLV does not satisfy the first constraint in Eq. (7), the MV can adjust the spacing by considering the CLV as its leading vehicle such that the constraint can be satisfied. In this case, it is unnecessary to calculate the desired spacing for the CLV using Eq. (10). Likewise, in case 1, there is no need to calculate the desired spacing for the CLV using Eq. (10) as both constraints of Eq. (7) are met. It is worth noting that the APS algorithm may yield a different pair of CVs for an MV at every time step, resulting in a different cooperative merging case. Such frequent updates of CVs and switching between different cases may cause traffic flow instability in the mainline. Fixing the pair of CVs and the merging case once determined for the first time may be inappropriate in the future since the traffic status changes over time. To address this, we introduce a decision time interval, denoted as T^{APS} , which determines how often the APS algorithm updates the pair of CVs and the merging case for each MV based on real-time traffic information every T^{APS} time. The pseudocode of the APS algorithm is summarized in Algorithm 1.

B. Collaborative Utility Choice (CUC) Model

In the proposed APS algorithm, a pair of CVs (i.e., a CLV and a CFV) and a cooperative merging case are assigned specifically for each MV every T^{APS} time. Once these are determined, it is necessary to control or provide suggestions to the CLV and CFV based on their collaborative attribute col to maintain a safe merging spacing for the MV.

Existing on-ramp merging models in the literature [28], [29], [30] are only applicable to scenarios where the main road has a single lane. In such cases, the CV can only accelerate or decelerate on the current lane to provide sufficient merging space for the MV. However, the impact of MV deceleration on upstream vehicles can take a long time to dissipate, which reduces the road's traffic capacity and increases the risk of rear-end collision between the CV and the vehicle behind it [31]. When the main road has more than one lane, such as two lanes, the CV has two choices to cooperate with the MV: 1) Change from lane 2 to lane 1; 2) Continue driving in lane 2. In this case, the methods mentioned in the literature [28], [29], [30] are no longer applicable. Therefore, we propose a CUC model for cooperative merging control in multilane freeway merging areas. The CUC model is a discrete choice model that enables the CV to make the best choice between maintaining longitudinal movement and changing lanes based on the utility. This way, the vehicles on the mainline are encouraged

Algorithm 1 Anticipatory Position Searching (APS) Algorithm

```

DEFINE  $t = 0$ 
 $g_{min}^{CM} = 1.2 \text{ s}$ 
FOR each merging vehicle (MV) within the on-ramp
  DEFINE  $t_{MV,APS}^0$  as the initial decision time of the APS
  algorithm of MV
  IF  $(t - t_{MV,APS}^0) \bmod T^{APS} \equiv 1$ 
    FOR each vehicle ( $i$ ) within lane 2
      IF  $x_i(t) \leq x_{MV}(t) + L^{cr}$  AND  $x_i(t) \geq x_{MV}(t) - L^{cr}$ 
         $i \in Gap_{car\ MV}(t)$ ;
      END IF
       $Gap_{car\ MV}(t) = [G_1, G_2 \dots, G_n]$ ;
    END FOR
     $T_{MV}^*(t) = \frac{x_0^m - x_{MV}(t)}{v_{MV}(t)}$ ;
    FOR each vehicle  $j \in Gap_{car\ MV}(t)$ 
       $x_j^*(t) = v_j(t) \cdot T_{MV}^*(t) + x_j(t)$ ;
       $\mathcal{D}_j^*(t) = x_j^*(t) - x_0^m - l$ ;
    END IF
     $\mathcal{D}_{MV}(t) = [\mathcal{D}_1^*(t), \mathcal{D}_2^*(t), \dots, \mathcal{D}_j^*(t)]$ ;
    IF  $\mathcal{D}_j^*(t) > 0$  AND  $\mathcal{D}_{j+1}^*(t) < 0$ 
       $j$  is the CLV AND the vehicle that follows vehicle
       $j$  is the CFV;
    END IF
     $\mathcal{D}_{min}^{CLV}(t) = v_{MV}(t) \cdot g_{min}^{CM}$ ;
     $\mathcal{D}_{min}^{CFV}(t) = v_{CFV}(t) \cdot g_{min}^{CM}$ ;
    IF  $\mathcal{D}_{CLV}^*(t) \geq \mathcal{D}_{min}^{CLV}(t)$  AND  $|\mathcal{D}_{CFV}^*(t)| \geq \mathcal{D}_{min}^{CFV}(t)$ 
      CASE 1:  $col_{CLV} = 0$  AND  $col_{CFV} = 0$ ;
    ELSE IF  $\mathcal{D}_{CLV}^*(t) \geq \mathcal{D}_{min}^{CLV}(t)$  AND  $|\mathcal{D}_{CFV}^*(t)| < \mathcal{D}_{min}^{CFV}(t)$ 
      CASE 2:  $col_{CLV} = 0$  AND  $col_{CFV} = 1$ ;
      SET the virtual vehicle MV';
    ELSE IF  $\mathcal{D}_{CLV}^*(t) < \mathcal{D}_{min}^{CLV}(t)$  AND  $|\mathcal{D}_{CFV}^*(t)| \geq \mathcal{D}_{min}^{CFV}(t)$ 
      CASE 3:  $col_{CLV} = 1$  AND  $col_{CFV} = 0$ ;
    ELSE IF  $\mathcal{D}_{CLV}^*(t) < \mathcal{D}_{min}^{CLV}(t)$  AND  $|\mathcal{D}_{CFV}^*(t)| < \mathcal{D}_{min}^{CFV}(t)$ 
      CASE 4:  $col_{CLV} = 1$  AND  $col_{CFV} = 1$ ;
      SET the virtual vehicle MV';
    END IF
  END IF
END FOR
 $t = t + time\_step$ ;

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to leave more space upstream of the on-ramp to reduce merging conflicts, improving the efficiency of the entire merging process. To highlight the different characteristics of CHVs and CAVs, we make the following statements.

- 1) Although CHVs can receive instructions determined by the CUC model, not all of them are willing to follow these instructions. Hence, the driver compliance rate is introduced to account for this phenomenon. Two types of CHVs are classified according to the compliance, namely the compliant and non-compliant CHVs.
- 2) The compliance rate C of CHVs is calculated as the percentage of compliant CHVs among all CHVs. When

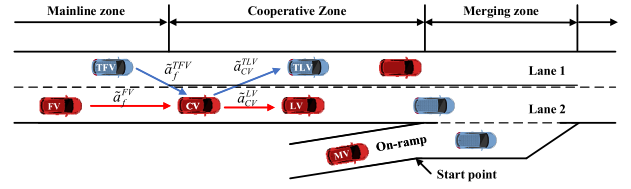


Fig. 4. Illustration of on-ramp cooperative system.

C equals to 0, all CHVs are non-compliant CHVs, which can also be referred to HVs. When C equals to 1, all CHVs are assumed to fully comply with the suggestions given by the CUC model.

- 3) The non-compliant CHVs do not consider the suggestions given by the CUC model and continue to drive in their original states, while the CAVs and compliant CHVs fully comply with the CUC model within the cooperative and merging zones. Although both CAVs and compliant CHVs adhere to the guidelines of the CUC model, they exhibit different longitudinal behaviors. As such, this study considers the distinction between CAVs and CHVs in mixed traffic flow from both the longitudinal and lateral behavior perspectives.

As shown in Fig. 4, the TLV and TFV represent the leading and following vehicles of the CV in the target lane (lane 1), respectively. The CV can be the CLV or CFV defined in section III-A. The FV and LV represent the leading and following vehicles of the CV in the current lane, respectively.

During the cooperative process, the CV faces two choices, described as follows:

Choice 1: The CV changes from lane 2 to lane 1, where the TFV and CV form a subsystem, and the CV and TLV form another subsystem. The new accelerations of the TFV and CV are denoted by $\tilde{a}_{TFV}^{CV}$ and $\tilde{a}_{CV}^{TLV}$, respectively. The TLV, CV, and TFV form a lane-changing demand system.

Choice 2: The CV continues to drive in lane 2, where the FV and CV form a subsystem, and the CV and LV form another subsystem. The new accelerations of the FV and CV are denoted by $\tilde{a}_{FV}^{CV}$ and $\tilde{a}_{CV}^{LV}$, respectively. The LV, CV and FV form a no lane-changing demand system.

The CUC model assumes that the CV considers not only its own safety but also the impact of each choice on the upstream vehicles, as well as the acceleration and comfort benefits that can be obtained through the choice during the process of collaborative choice making. Therefore, the CV makes the choice with higher overall system benefits.

The utility functions of the two choices are given by:

$$U_1(t) = \alpha \cdot (c_{CV}^{TLV}(t) + c_{TFV}^{CV}(t)) + \beta \cdot \tilde{a}_{TFV}^{CV}(t) + \gamma \cdot \tilde{a}_{CV}^{TLV}(t) + \zeta \cdot \left| \tilde{a}_{CV}^{TLV}(t) - a_{CV}^{LV}(t) \right| \quad (11)$$

$$U_2(t) = \alpha \cdot (c_{CV}^{LV}(t) + c_{FV}^{CV}(t)) + \beta \cdot \tilde{a}_{FV}^{CV}(t) + \gamma \cdot \tilde{a}_{CV}^{LV}(t) + \zeta \cdot \left| \tilde{a}_{CV}^{LV}(t) - a_{CV}^{LV}(t) \right| \quad (12)$$

$$c_f^l(t) = \frac{(v_f(t) - v_l(t))^2}{x_l - x_f - L_l} \quad (13)$$

where $U_1(t)$ and $U_2(t)$ represent the utilities of the two choices available to the CV at time t , respectively. The acceleration of

the CV at time t is denoted by $a_{CV}^{LV}(t)$. $c_f^l(t)$ is the deceleration rate that vehicle f needs to be taken to avoid crash relative to vehicle l at time t according to [32], [33], and [34], which serves as a safety measure index. Here, $l \in \{CV, LV, TLV\}$ and $f \in \{CV, TFV, FV\}$. The smaller the value of $c_f^l(t)$, the safer it is. α , β , γ , and ζ are weight coefficients, where $\alpha < 0$, $\beta > 0$, $\gamma > 0$ and $\zeta < 0$. The first terms of Eqs. (11) and (12) denote the safety considerations of the CV and the following vehicle, with a larger value indicating greater safety. The second terms reflect the degree of influence on the following vehicle, with a smaller value indicating a higher impact. The third terms denote the acceleration benefits that the CV can obtain from its choices, with a higher value indicating a higher potential speed. The fourth terms represent the difference between the current and new accelerations of the CV, characterizing the driving comfort. The smaller the value of the fourth term, the more comfortable it is.

When the CV chooses choice 2, the following safety constraints need to be considered:

$$\begin{cases} TT_{CV}^{TLV}(t) \geq TT_{min} \\ TT_{TFV}^{CV}(t) \geq TT_{min} \end{cases} \quad (14)$$

with

$$TT_f^l(t) = \frac{-\Delta V_f(t) \mp \sqrt{\Delta V_f(t)^2 + 2\Delta a_f(t)d_f(t)}}{\Delta a_f(t)}, \quad (15)$$

$l \in \{TLV, CV\}, f \in \{CV, TFV\}$

where TT is the time that a following vehicle would take to collide with a leading vehicle if both vehicles maintain their current speed and acceleration [32], [33]. Generally, the value of TT below 1.5s is unsafe. Therefore, the value of TT_{min} is set to be 1.5s [33]. The final TT value is obtained as follows: 1) If both TT values are positive, the smaller one is chosen as the final value; 2) If one TT value is positive and the other is negative, the positive one is chosen as the final value.

Let M_{CV} denote the choice of the CV, the cooperative decision of the CV can be obtained as:

$$M_{CV} = \begin{cases} \text{Choice1, } U_1(t) > U_2(t) \text{ and Eq. (14) is satisfied} \\ \text{Choice2, } U_1(t) \leq U_2(t) \end{cases} \quad (16)$$

IV. LOWER LAYER OF CORMC MODEL

A. Longitudinal and Lane Changing Models

1) Longitudinal Models:

a) *Longitudinal model of CAVs:* We consider a subsystem consisting of two adjacent vehicles, in which the following vehicle i performs longitudinal movement and follows the preceding vehicle $i-1$ safely according to the car following model. Let $x_i(t)$, $v_i(t)$, $S_i(t)$, $d_i(t)$ and $C_i(t)$ denote the longitudinal position, speed, desired spacing, actual spacing and collision avoidance spacing of vehicle i at time t , respectively. The kinematic relationship between the two vehicles in the subsystem is shown in Fig. 5, we can obtain:

$$d_i(t) = x_{i-1}(t) - x_i(t) - L_{i-1} \quad (17)$$

$$S_i(t) = v_i(t)h_i + d_0 + C_i(t) \quad (18)$$

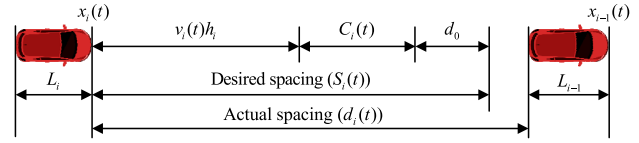


Fig. 5. Illustration of the kinematic relationship between vehicles in the subsystem.

where $x_{i-1}(t)$ represents the longitudinal position of vehicle $i-1$ at time t . L_{i-1} and L_i represent the length of vehicle $i-1$ and vehicle i , respectively. h_i represents the time gap of vehicle i . d_0 is the stopping spacing.

As for the CAV's car-following model, the collision avoidance spacing $C_i(t)$ is given by:

$$C_i(t) = \begin{cases} \frac{(v_i(t) - v_{i-1}(t))^2}{2|a_{min}|}, & v_i(t) > v_{i-1}(t) \\ 0, & \text{others} \end{cases} \quad (19)$$

where a_{min} represents the minimum acceleration of vehicle i . To ensure safety, the speed of vehicle i must be less than or equal to the speed of vehicle $i-1$. If the speed of vehicle i is higher than the speed of vehicle $i-1$, a collision may occur. Therefore, we introduce a collision avoidance spacing for the desired spacing, which is essentially the distance traveled by vehicle i when it decelerates at the maximum deceleration until its speed equals that of vehicle $i-1$. The larger the speed difference between the two vehicles, the larger the collision avoidance spacing is required.

To perform the longitudinal control of CAVs, we consider two possible driving modes.

- 1) **The cruising mode**, under which the CAV can maintain the desired speed when the preceding vehicle is absent or the actual time gap is more than two times of the desired time gap [3].
- 2) **The gap-regulating mode**, which is used to maintain a constant time gap of the CAV with its preceding vehicle under the car-following condition.

In the cruising mode, we can obtain the vehicle acceleration as:

$$a_i(t) = k_1(v_e - v_i(t)) \quad (20)$$

where k_1 represents the control gain parameter, and the value of k_1 is set to be $0.4s^{-1}$ according to [35]. v_e represents the equilibrium speed. $a_i(t)$ and $v_i(t)$ represent the acceleration and the velocity of vehicle i at time t , respectively.

In the gap-regulating mode, we use the Cascade Proportional Integral Derivate (CPID) model, which is an acceleration model, to control the longitudinal maneuver of CAVs. The CPID model combines an inner loop controlling the velocity error and an outer loop controlling the spacing error with objectives of reducing both errors to zero. In our previous work, we demonstrated the strong stability and good performance of the CPID model [27]. To account for the heterogeneity of CAVs, we introduce a random variable $\tau(\omega)$ to represent the inertial lag. We define the random parameter $\phi(\omega) \equiv \tau(\omega)$ with distribution function F_ϕ . The inertial lag τ_i for each CAV i is drawn independently from this common

distribution: $\phi_i \sim F_\phi$. The mathematical formulations of the stochastic CPID model are given by:

$$ex_i(t) = d_i(t) - S_i(t) \quad (21)$$

$$ev_i(t) = v_{i-1}(t) - v_i(t) \quad (22)$$

$$\begin{aligned} \partial_i(t) = & K_{px} ex_i(t) + K_{ix} \int_0^t ex_i(\xi) d\xi \\ & + K_{dx} \frac{d}{dt} ex_i(t) \end{aligned} \quad (23)$$

$$e_i(t) = \partial_i(t) - ev_i(t) \quad (24)$$

$$u_i(t) = K_{pv} e_i(t) + K_{iv} \int_0^t e_i(\xi) d\xi + K_{dv} \frac{d}{dt} e_i(t) \quad (25)$$

$$a_i(t + \Delta t, \omega) = \left(1 - \frac{\Delta t}{\tau_i(\omega)}\right) a_i(t) + \frac{\Delta t}{\tau_i(\omega)} u_i(t) \quad (26)$$

$$u_{min} \leq u_i(t) \leq u_{max}$$

$$a_{min} \leq a_i(t) \leq a_{max}$$

$$v_{min} \leq v_i(t) \leq v_{max} \quad (27)$$

where, ω is used to distinguish between the stochastic relations and the deterministic ones, e.g., $\tau_i(\omega)$ and $\tau_i(\cdot)$. $ex_i(t)$ is the spacing error of vehicle i compared with the desired spacing at time t . $ev_i(t)$ is the speed error of vehicle i relative to vehicle $i-1$ at time t . $\partial_i(t)$ and $u_i(t)$ represent the output of the outer and inner loop PIDs of the cascade PID model at time t , respectively. e_i represents the input error of the inner loop PID. ξ is the integral variable. K_{px} , K_{ix} and K_{dx} are the parameters of the outer loop PID. K_{pv} , K_{iv} and K_{dv} are the parameters of the inner loop PID. Δt represents the time interval. u_{min} and u_{max} are the maximum and minimum constraints on u_i , respectively. $a_i(t)$ represents the acceleration of vehicle i at time t . a_{min} and a_{max} are the maximum and minimum constraints on a_i . v_{min} and v_{max} are the maximum and minimum constraints on v_i .

b) Longitudinal model of HVs: We utilize the Intelligent Driver Model (IDM) developed by Treiber et al. [36] to describe the longitudinal behavior of CHVs. The IDM model provides greater realism in capturing congested dynamics [37], and produce collision-free and smooth traffic flow [5]. To account for the heterogeneity of CHVs, we assume that v^f is a random variable following a normal distribution. We define the random parameter $\vartheta(\omega) \equiv v^f(\omega)$ with distribution function F_ϑ , and the desired speed v_i^f for each HV i is drawn independently from this common distribution: $\vartheta_i \sim F_\vartheta$. The mathematical formulation of the stochastic IDM is given by:

$$a_i(t, \omega) = A_i \left[1 - \left(\frac{v_i(t)}{v_i^f(\omega)} \right)^4 - \left(\frac{S_i^*(t)}{d_i(t)} \right)^2 \right] \quad (28)$$

$$S_i^*(t) = d_0 + \max(0, v_i(t) h_i - \frac{v_i(t) [v_{i-1}(t) - v_i(t)]}{2\sqrt{A_i b_i}}) \quad (29)$$

where, $v_i^f(\omega)$ is the random parameter describing the desired speed of CHV i . A_i , b_i represent the maximum and comfort decelerations of vehicle i , respectively. d_0 represents the stopping distance. $a_i(t)$, $v_i(t)$, $S_i^*(t)$, $d_i(t)$ represent the

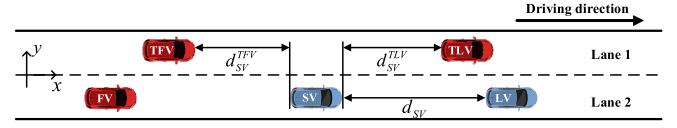


Fig. 6. Illustration of lane-change spacing for vehicles.

acceleration, speed, desired car-following distance, and actual car-following distance of vehicle i at time t , respectively. $v_{i-1}(t)$ is the speed of vehicle $i-1$ at time t . h_i represents the desired time gap of vehicle i .

2) Lane-Changing Model: In this section, we provide detailed modeling of the lateral movement process, including the lane-changing process of vehicles in the mainline and the merging process of vehicles in the on-ramp, which can be considered as a specific type of lane-changing. The whole process is composed of three parts: 1) lane-changing (or merging) decision-making, 2) dynamic lane-changing trajectory planning, and 3) trajectory tracking control. It is worth noting that the lane-changing model in this section describes the lateral movement characteristics of both CAVs and CHVs.

a) Lane-changing decision: It is worth noting that the lane-changing decision of vehicles in the main lanes has different constraints within different zones defined in Fig. 1. In this subsection, we discuss only the lane-changing decision within the mainline zone. The lane-changing decision within the cooperative and merging zones is discussed in sections III-B and IV-B. As shown in Fig. 6, d_{SV} , d_{SV}^{TLV} and d_{TFV}^{SV} denote the longitudinal spacing between the subject vehicle (SV) and LV, the longitudinal spacing between the SV and TLV, and the longitudinal spacing between the TFLV and SV, respectively. To simplify the lane-changing decision-making behavior of vehicles within the mainline, we consider rule-based lane-changing criteria, which include the speed constraint and two safety spacing constraints given by:

$$v_{TLV} - v_{SV} \geq \Delta v_{threshold} \quad (30)$$

$$d_{TFV}^{SV}(t) \geq v_{TFV}(t) \cdot h_{min}^{LC} \quad (31)$$

$$d_{SV}^{TLV}(t) > v_{SV}(t) \cdot h_{min}^{LC} \quad (32)$$

where h_{min}^{LC} represents the minimum acceptable time gap for lane-changing. Eq. (30) indicates that if the SV's speed is constrained by the LV such that the SV can't achieve a faster velocity, then the SV chooses to change lanes.

When the above lane-changing criteria are met, the following changes occur: 1) The SV and TLV form a new subsystem, with the TLV being considered as the new leader; 2) The TFLV treats the SV as its new leader and they constitute a new subsystem. The TFLV performs cooperative maneuvers based on the state of the SV, ensuring the safety of the lane-changing process; 3) At the meantime, the FV still regards the SV as its leader, and they maneuver cooperatively as a subsystem.

b) Dynamic lane-changing trajectory planning: We use an improved sine function for the lane-changing trajectory planning. The good performance of this method has been demonstrated in our previous work [27]. The mathematical description of the lane-changing trajectory planning model is

given by:

$$y_{SV}^r(x_{SV}^r(t)) = y_{SV}^0 + \frac{y_d^0}{2\pi} \left\{ \frac{2\pi}{M(t)} (x_{SV}^r(t) - x_{SV}^0) \right\} - \sin \left[\frac{2\pi}{M(t)} (x_{SV}^r(t) - x_{SV}^0) \right] \\ x_{SV}^r(t) \in [x_{SV}^0, x_{SV}^0 + M(t)] \quad (33)$$

$$M(t) = v_{SV}(t) \sqrt{\frac{2|y_d^0|}{a_p}} + L_{centerline} \quad (34)$$

$$v_{SV}(t) = v_{SV}(t - \Delta t) + a_{SV}(t) \cdot \Delta t \quad (35)$$

$$y_d^0 = y_{TLV}^0 - y_{SV}^0 \quad (36)$$

$$y_{SV'}^r(x_{SV}^r(t)) = \frac{y_d^0}{M(t)} \left\{ 1 - \cos \left[\frac{2\pi}{M(t)} (x_{SV}^r(t) - x_{SV}^0) \right] \right\} \quad (37)$$

$$y_{SV''}^r(x_{SV}^r(t)) = \frac{\pi a_p}{v_{SV}^2(t)} \sin \left[\frac{2\pi}{M(t)} (x_{SV}^r(t) - x_{SV}^0) \right] \quad (38)$$

$$K(x_{SV}^r(t)) = \frac{y_{SV''}^r(x_{SV}^r(t))}{\left[1 + (y_{SV'}^r(x_{SV}^r(t)))^2 \right]^{\frac{2}{3}}} \quad (39)$$

$$\varphi_{SV}^r(x_{SV}^r(t)) = \tan^{-1}(y_{SV'}^r(x_{SV}^r(t))) \quad (40)$$

$$\delta_{SV}^r(x_{SV}^r(t)) = \tan^{-1}(l * K(x_{SV}^r(t))) \quad (41)$$

where $x_{SV}^r(t)$ and $y_{SV}^r(t)$ represent the planned longitudinal and lateral locations of the SV at time t , respectively. y_d^0 is the lateral distance between the TLV and SV at the beginning of the lane-changing maneuver. x_{SV}^0 , y_{SV}^0 and y_{TLV}^0 represent the longitudinal position, lateral position of the SV and lateral position of the TLV at the beginning of the lane-changing process, respectively. $M(t)$ represents the longitudinal length of the planned trajectory at time t ; $L_{centerline}$ is the distance that the SV will continue to track the centerline of the lane after completing a lane change. The parameter is designed to prevent errors during the lane-changing process that could cause the SV to have the non-zero yaw and front wheel angles when it reaches the centerline of the target lane. In this study, we set $L_{centerline}$ to 100m. Notably, the lane change time can

be approximated using $\sqrt{\frac{2|y_d^0|}{a_p}} \cdot v_{SV}(t)$ represents the velocity of the SV at time t , which is the safe lane-changing speed derived based on the longitudinal model. a_p is the planned acceleration considering the comfort of lane changing. $y_{SV'}^r$ and $y_{SV''}^r$ represent the first derivative and second derivative of y_{SV}^r , respectively; K is the curvature of y_{SV}^r ; φ_{SV}^r and δ_{SV}^r represent the reference yaw angle and reference front wheel angle of the SV, respectively; l is the distance between the front and rear axles of the SV.

c) *Front-collision-avoidance algorithm*: During a lane-changing process, it is crucial that the SV avoids potential collisions with the TLV, TFLV and LV. The first two collision-avoidance are determined by longitudinal models. However, the SV may also be at risk of collision with the LV prior to changing lanes. To address this issue, a front-collision-avoidance algorithm is developed for lane changing process, aiming at preventing collisions between the SV and LV. Let $T_{mid}(t)$ denote the time required for the SV to travel from the

current position to the midpoint of the planned trajectory, given its current speed $v_{SV}(t)$. Let $d_{mid}(t)$ denote the difference between the longitudinal displacement of the SV and that of the LV after they have each traveled for time $T_{mid}(t)$ at their respective current speeds. We can obtain:

$$x_{SV}^{mid}(t) = x_{SV}^0 + \frac{M(t)}{2} \quad (42)$$

$$T_{mid}(t) = \frac{x_{SV}^{mid}(t) - x_{SV}(t)}{v_{SV}(t)} \quad (43)$$

$$d_{mid}(t) = T_{mid}(t) \cdot (v_{SV}(t) - v_{LV}(t)) \quad (44)$$

where $x_{SV}^{mid}(t)$ represents the longitudinal position of the midpoint of the SV's planned trajectory at time t . The mechanism of collision-avoidance is to ensure that the SV does not collide with the LV when the SV travels to the midpoint of its planned trajectory (lane dividing line) at each time step, during which the speeds of the SV and LV are assumed to be constant. That is, at each time step, the longitudinal gap spacing between the SV and LV at time t is larger than $d_{mid}(t)$. Therefore, the collision-avoidance constraint can be obtained as:

$$d_{mid}(t) < x_{LV}(t) - x_{SV}(t) - L_{LV} \quad (45)$$

By substituting Eqs. (34) and (42)-(44) into Eq. (45), we can obtain:

$$\frac{x_{SV}^0 + \frac{1}{2} \cdot v_{SV}(t) \sqrt{\frac{2L_w}{a_p}} - x_{SV}(t)}{v_{SV}(t)} \cdot (v_{SV}(t) - v_{LV}(t)) < x_{LV}(t) - x_{SV}(t) - L_{LV} \quad (46)$$

If the above constraint is unsatisfied, $v_{SV}(t)$ used in the trajectory planning is equal to $v_{SV}(t - 1)$, which is the speed of the SV at the previous time.

d) *Trajectory tracking control*: In this study, we use the MPC controller to perform the real-time lateral tracking control of the planned trajectory during the lane changing and merging processes. We denote $\chi = [x, y, \varphi]$ as the system state vector, which includes the longitudinal location x , lateral position y , yaw angle φ . The discrete state-space equation is given by:

$$\bar{\chi}(k+1) = A(k) \bar{\chi}(k) + B(k) \bar{u}(k) \quad (47)$$

with

$$A(k) = \begin{bmatrix} 1 & 0 & -v \sin \varphi_r \Delta t \\ 0 & 1 & v \cos \varphi_r \Delta t \\ 0 & 0 & 1 \end{bmatrix} \quad B(k) = \begin{bmatrix} 0 \\ 0 \\ \frac{v \Delta t}{l \cos^2 \delta_{fr}} \end{bmatrix} \quad (48)$$

$$\bar{\chi}(k) = \begin{bmatrix} x(k) - x_r(k) \\ y(k) - y_r(k) \\ \varphi(k) - \varphi_r(k) \end{bmatrix} \quad \bar{u}(k) = (u(k) - u_r(k)) \quad (49)$$

where, the system control variable u is equal to δ_f , which is the front wheel angle. l is the distance between the front and the rear axles, and Δt is the time step.

To ensure that the vehicle can track the reference trajectory quickly and smoothly, the objective function is formulated as:

$$\min J(k) = \sum_{i=1}^{n_p} \|\eta(k+i|k) - \eta_{ref}(k+i|k)\|_Q + \sum_{i=1}^{N_c-1} \|\Delta U(k+j|k)\|_n^2 + \rho \varepsilon^2 \quad (50)$$

with constraints:

$$\begin{cases} \Delta U_{min} \leq \Delta U(k) \leq \Delta U_{max} \\ U_{min} \leq A\Delta U(k) + U(k-1) \leq U_{max} \end{cases} \quad (51)$$

where N_p and N_c represent the prediction and control horizons, respectively. $\eta(k+i|k)$ represents the vehicle state at prediction step i when k is treated as the current sampling point. η_{ref} represents the reference state obtained from the planned trajectory. Q and R are the weight matrices. $\Delta U(k)$ is the system control variable increment at time step k . ρ is the weight coefficient. ε is the relaxation factor. ΔU_{min} , ΔU_{max} are the limits of the control variable increment. A is the coefficient matrix of the constraint equation.

The trajectory tracking control is then converted into a Quadratic Programming (QP) problem under the MPC framework. The good performance of the MPC tracking controller has been proved in our previous work [27].

B. Cooperative Merging Control (CMC) Model

The CMC model includes the dynamic time gap acceptance model, merging behavior model and boundary-collision-avoidance algorithm. During the merging process, the vehicle's merging intention is higher as it approaches the downstream boundary of the on-ramp, leading to a smaller acceptable merging time gap. To characterize this merging decision behavior, we introduce the dynamic time gap acceptance model that determines the minimum acceptable time gap for cooperative merging. The MV assesses the merging gap in the target lane and determines the feasibility of a merging maneuver using Eq. (53). The merging behavior model, which includes merging decision-making, merging trajectory planning, and merging trajectory tracking control, iteratively controls the MV for each time step until it reaches the centerline of the target lane. The boundary-collision-avoidance algorithm ensures collision-free between MVs and the downstream boundary of the on-ramp by deriving the speed constraints of the MV.

1) *Dynamic Time Gap Acceptance Model*: Because merging is a mandatory lane change due to the strong restriction of the downstream boundary of the ramp, we assume that MV's minimum acceptable time gap for cooperative merging gradually decreases as the distance between the MV and downstream boundary of the ramp becomes smaller. This is consistent with actual merging decision in traffic. Therefore, the dynamic time gap acceptance model is formulated as:

$$\tilde{h}_{MV}^{CM}(t) = h_{upper}^{CM} (1 - \zeta \frac{x_{MV}(t) - x_0^m}{(x_{ramp}^{end} - x_0^m)}) \quad (52)$$

where $\tilde{h}_{MV}^{CM}(t)$ represents the MV's minimum acceptable time gap of cooperative merging at time t . h_{upper}^{CM} represents the upper bound of the acceptable time gap of cooperative merging. x_0^m represents the longitudinal position of the start point of the merging zone. x_{ramp}^{end} represents the longitudinal position of the downstream boundary of the on-ramp. ζ represents the linear coefficient of the dynamic time gap acceptance model. Eq. (52) indicates that as the distance between the MV and the start point of the merging zone increases, $\tilde{h}_{MV}^{CM}(t)$ decreases

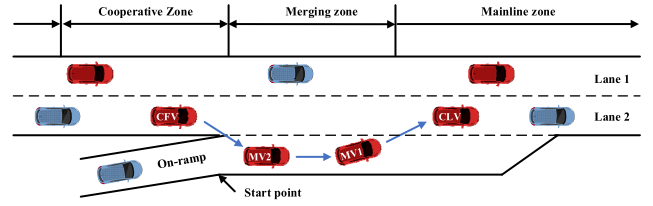


Fig. 7. Illustration of the scenario where multiple MVs merge to the adjacent main lane (lane 2).

linearly. If the MV is at the start point of the merging zone, $\tilde{h}_{MV}^{CM} = h_{upper}^{CM}$, and if the MV is at the downstream boundary of the ramp, $\tilde{h}_{MV}^{CM}$ equals to $(1 - \xi)h_{upper}^{CM}$.

2) *Merging Behavior Model*: The merging behavior model is to describe the merging process of the MV, which is essentially a process of lane-changing from the on-ramp to the adjacent lane (lane 2). Therefore, we can adopt the lane-changing model proposed in section IV-A.2.) to describe the MV's merging behavior. It is noteworthy that the merging decision making is different from the lane-changing decision. The MV starts to make merging maneuver when the follow constraints are satisfied:

$$\begin{cases} d_{CFV}^{MV}(t) \geq v_{CFV}(t) \cdot \tilde{h}_{MV}^{CM}(t) \\ d_{MV}^{CLV}(t) \geq v_{MV}(t) \cdot \tilde{h}_{MV}^{CM}(t) \end{cases} \quad (53)$$

where $d_{CFV}^{MV}(t)$ represents the actual spacing between the CFV and MV at time t . $d_{MV}^{CLV}(t)$ represents the actual spacing between the MV and CLV at time t . $v_{CFV}(t)$ and $v_{MV}(t)$ represent the speeds of the CFV and MV at time t , respectively.

The merging trajectory planning and tracking control methods of MVs are the same as those in the lane changing model, which is provided in section IV-A.2).

Collaborative merging involving multiple vehicles is a common in real-world traffic, as depicted in Fig. 7. In this situation, the MVs can be viewed as a vehicle platoon. In [38], two vehicles are simply regarded as one vehicle by adding their lengths together, which is impractical. To address this limitation, we propose the following rules for the merging process:

- 1) Only the first and last MVs in the platoon need to be considered in the merging process. Specifically, the first MV and the CLV form a new subsystem, while the CFV and the last MV constitute another subsystem.
- 2) The remaining MVs in the platoon continue to maintain car-following in the current subsystem. In other words, the speed of these MVs required for merging (lane-changing) trajectory planning in Eqs. (33) and (34) is updated within the current subsystem.

Under these rules, the CFV only needs to maintain a safe spacing with the last vehicle in the platoon, which is consistent with the real-world multi-vehicle merging scenarios. The leading vehicle of the platoon ensures the safety of the entire platoon merging process.

3) *Boundary-Collision-Avoidance Algorithm*: Boundary-collision-avoidance is crucial to ensure collision-free merging between MVs and the downstream boundary of the on-ramp. The mechanism of collision-avoidance is shown in Fig. 8.

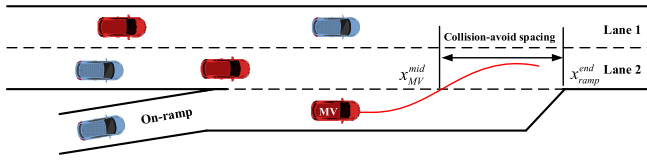


Fig. 8. Illustration of the boundary-collision-avoidance for downstream ramp of MVs.

$x_{MV}^{mid}(t)$ represents the longitudinal position of the midpoint of the MV's planned trajectory at time t . x_{ramp}^{end} is the longitudinal position of the downstream boundary of the on-ramp. Merging is a specific type of lane-changing, so the collision-avoidance spacing requirement must be satisfied when the center of the MV travels to the midpoint of its planned trajectory at each time step. Here, the collision-avoidance spacing is the stopping distance d_0 . The boundary-collision-avoidance condition can be expressed as:

$$x_{ramp}^{end} - x_{MV}^{mid}(t) - \frac{L_{MV}}{2} > d_0 \quad (54)$$

$$x_{MV}^{mid}(t) = x_{MV}(t) + \frac{1}{2}v_{MV}(t)\sqrt{\frac{2L_w}{a_p}} \quad (55)$$

where $x_{MV}(t)$ represents the longitudinal position of the MV at time t . L_{MV} is the length of the MV. L_w is the lane width. a_p represents the planned acceleration in trajectory planning. By substituting Eq. (55) into Eq. (54), we can obtain:

$$v_{MV}(t) < (x_{ramp}^{end} - x_{MV}(t) - \frac{L_{MV}}{2} - d_0) \cdot \sqrt{\frac{2a_p}{L_w}} \quad (56)$$

Therefore, to avoid a boundary-collision, the speed of the MV at time t should satisfy the condition defined in Eq. (56).

V. NUMERICAL EXPERIMENTS

A. Vehicle Generation Model

In order to ensure continuous traffic flow on a simulated road section, a vehicle generation model is developed, which consists of two components: initialization and boundary generation. The initialization stage generates vehicles based on the density and road length prior to the start of the simulation, while the boundary generation stage produces upcoming vehicles at the upstream boundary of the road section during the simulation process, considering both traffic demand and the arrival headway distribution between two adjacent vehicles.

The theoretical capacity of mixed traffic can be calculated:

$$C_{mix}(P_1) = \frac{3600}{P_1 \cdot h_{CAV} + (1 - P_1) \cdot h_{CHV} + \frac{d_0 + L}{v_e}} \quad (57)$$

where C_{mix} represents the theoretical capacity (veh/h/lane) of mixed traffic. P_1 represents the penetration rate of CAVs. h_{CAV} and h_{CHV} represent the desired time gap of CAVs and CHVs, respectively. v_e represents the equilibrium speed. d_0 and L respectively represent the stopping spacing and length of vehicle. The traffic volume of the on-ramp lane Q_{ramp} is calculated as:

$$Q_{ramp} = C_{mix}(P_1) \cdot \eta_{ramp} \quad (58)$$

where η_{ramp} is a parameter representing the ratio of the on-ramp flow to the theoretical capacity of the mixed traffic.

In the vehicle generation model, we consider randomness and uncertainty of traffic flow from four aspects: the vehicle's arrival time headway, CHV's desired speed, CHV's compliance and CAV's inertial lag. Before the simulation starts, a set of temporary vehicles is pre-generated on each lane based on the input flow and the CAV penetration rate. For each vehicle, the vehicle type (CAV or CHV), desired time gap, driver compliance, desired speed (for CHVs) or inertial lag (for CAVs), lane attributes, and arrival time headway are assigned. Once the simulation starts, the system checks at each time step whether the headway of the waiting vehicle relative to its preceding vehicle (H_W) is greater than or equal to its assigned arrival time headway (H_A). If H_W is greater than H_A , the vehicle is generated.

In the previous studies, randomly distributed inertial lag for CAVs and a power-law headway distribution for the initial conditions are utilized [39], [40]. However, the arrival time headway distribution described by the negative exponential distribution is inconsistent with actual traffic flow characteristics, as it allows for extremely small headways (e.g., 0.01s), which is not conducive to safe driving. Therefore, we use the shifted negative exponential distribution to describe the headway distribution. The probability density function is given by:

$$f(h_{AT}) = \begin{cases} \frac{1}{1/\lambda - h_{shifted}} e^{-\frac{1}{1/\lambda - h_{shifted}}(h_{AT} - h_{shifted})}, & h_{AT} > 0 \\ 0, & h_{AT} \leq 0 \end{cases} \quad (59)$$

$$\lambda = Q/3600 \quad (60)$$

where Q is the traffic volume [veh/h/ln]. λ is the vehicle arrival rate per unit time [veh/s]. $h_{shifted}$ is the shifted time headway which is the minimum arrival time headway. h_{AT} is the arrival time headway.

B. Simulation Platform

To evaluate the effectiveness of the proposed algorithm and generate the required continuous traffic flow in a mixed traffic environment, we have developed a microscopic simulation platform using MATLAB. This platform incorporates the longitudinal and lane-changing models developed in sections IV-A, explicitly considering both longitudinal and lateral movements of vehicles. Additionally, the vehicle generation model proposed in section V-A is used to simulate vehicle arrival, taking into account of randomness and uncertainty. Fig. 9 shows the flow chart of the simulation process.

C. Scenario Settings and Parameters

To test the proposed CORMC model, a typical freeway segment with two main lanes in one direction and an on-ramp is created in the simulation platform, as shown in Fig 10. The mainline section has a total length of 10km, with the on-ramp located at 6950m downstream from the starting boundary of the mainline. The merging zone has a length of 300m,

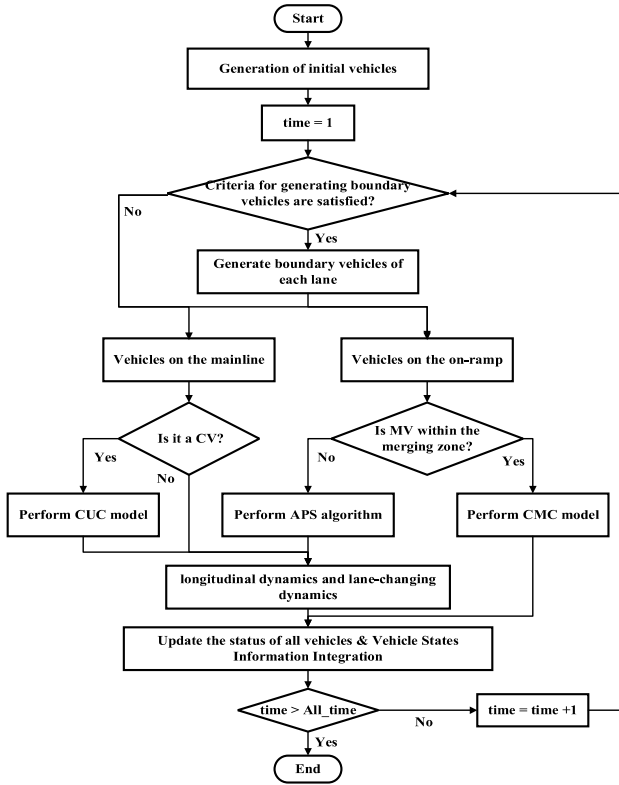


Fig. 9. The flow chart of the simulation process.

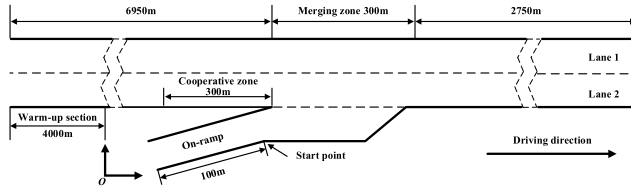


Fig. 10. Illustration of a simple merging network in simulation.

while the cooperative zone has a length of 300 m and moves dynamically according to the position of the MVs on the ramp, which is 300m upstream of each MV. Here, we assume that the communication range L^{cr} of the vehicle is 300m [41].

We set the penetration rate of CAVs to be 0%, 20%, 40%, 60%, 80% and 100% to investigate the impact of the CAV on the efficiency of the mainline traffic flow. To better understand the impact of different levels of merging flow ratios on the mainline traffic efficiency, we set the on-ramp flow ratios η_{ramp} between 10% and 40% with a 10% increment, while the mainline demand η_{main} is set to be 70% of the theoretical capacity. In this way, we can observe different traffic flow conditions (e.g., uncongested and congested) in the merging area when the on-ramp demand increases for each CAV penetration rate. A total of 24 scenarios are designed, and for each scenario, the simulation time period is 1200s. We use different random number seeds to generate simulation results, and take the average value of 20 simulations to reduce the randomness of the simulation experiment. The random seed assigns the vehicle type (CAV or CHV), the inertial lag of the CAV, the compliance and desired speed of the CHV, the initial speed, and the initial arrival time headway between two

TABLE I
THE VALUES OF VARIABLES AND PARAMETERS

	Parameter	Notation	Typical value
CAV's longitudinal model	Control gain parameter of the cruising mode	k_1	$0.4s^{-1}$
	Time interval	Δ	0.1 s
	Time inertial lag	τ_i	U (0.4, 0.7)
	Desired time gap of the CAV	h_{CAV}	1.2 s
	Minimum control value	u_{min}	-6 m/s ²
	Maximum control value	u_{max}	4 m/s ²
	Minimum acceleration	a_{min}	-6 m/s ²
IDM model	Maximum acceleration	a_{max}	4 m/s ²
	Desired time gap of the CHV	h_{CHV}	2 s
	Stopping spacing	d_0	2 m
	Vehicle length	L	4 m
	Desired speed	v_i^f	N (30, 1.5) m/s
	Maximum acceleration	A_i	1.25 m/s ²
	Comfort deceleration	b_i	2.09 m/s ²
Lane-changing model	Minimum acceptable time gap of lane-changing	h_{min}^{LC}	1.2 s
	Speed difference threshold for lane-changing	$\Delta v_{threshold}$	4 m/s
	Planned acceleration in trajectory planning process	a_p	0.1 m/s ²
APS algorithm	Anticipatory minimum acceptable time gap of merging	g_{min}^{CM}	1.2 s
	Decision time interval of APS algorithm	T^{APS}	5 s
	Length of the communication range	L^{cr}	300 m
CMC and CUC models	upper bound of the acceptable time gap of merging	h_{upper}^{CM}	1.2 s
	Linear coefficient of dynamic time gap acceptance model	ξ	2/3
	CUC model's weight coefficient	α	-1
	CUC model's weight coefficient	β	1.5
	CUC model's weight coefficient	γ	0.5
Vehicle generation model	CUC model's weight coefficient	ζ	-0.5
	Shifted time headway of lane 1, lane 2 and on-ramp	$h_{shifted}$	1.2 s / 1.2 s / 3.5 s
	Equilibrium speed	v_e	30 m/s
	Maximum speed of lane 1, lane 2 and on-ramp	v_{max}	33 m/s / 33 m/s / 30 m/s
	Initial speeds of vehicles in lane 1, lane 2 and on-ramp	v	30 m/s / 30 m/s / 16 m/s
	Simulation time period	All_time	1200 s

adjacent vehicles in the vehicle generation model. The values of the model variables and parameters are listed in Table I.

VI. RESULTS

A. Assessment Indicators of Traffic Efficiency

To evaluate the impact of the proposed CORMC model on traffic efficiency, we used the average flow and average speed defined by Edie based on the time-space trajectories of vehicles. Edie's generalized average flow and average speed are given by:

$$k(A) = \frac{t(A)}{|A|} \quad (61)$$

$$q(A) = \frac{d(A)}{|A|} \quad (62)$$

$$v(A) = \frac{d(A)}{t(A)} = \frac{q(A)}{k(A)} \quad (63)$$

where A represents the closed region observed on the time-space trajectory diagram. The area of the region is denoted by $|A|$, and $q(A)$, $k(A)$ and $v(A)$ represent Edie's

TABLE II

MAINLINE TRAFFIC EFFICIENCY WITH DIFFERENT COMPLIANCE RATES

		Compliance rates C (%)				
		0%	25%	50%	75%	100%
Flow (veh/h/ lane)	Merging flow	1389	1373	1430	1469	1521
	Diff (%)	-	-1.2	3.0	5.8	9.5
	Mainline flow	1553	1556	1567	1616	1699
	Diff (%)	-	0.2	0.9	4.1	9.4
Speed (m/s)	Merging speed	20.8	21.1	20.5	21.4	22.1
	Diff (%)	-	1.4	-1.4	2.9	6.3
	Mainline speed	24.3	24.8	25.2	26.1	26.4
	Diff (%)	-	2.1	3.7	7.4	8.6

generalized flow, average density and average speed in the observed time-space region A , respectively. The total time spent by all vehicles in region A is denoted by $t(A)$, while the total distance travelled by all vehicles in region A is denoted by $d(A)$.

In order to investigate the impact of ramp merging on mainline traffic, we divided the section between 4 km and 10 km downstream of the starting boundary of the mainline into 12 observation regions, with each region covering a distance of 500m. We then calculated the average flow and speed of each observation region during the total simulation time.

B. Impact of CHV Compliance Rate on CORMC Model

To better reflect the impact of the compliance rate on the effectiveness of the proposed CORMC model and its effects on mixed traffic performance, we analyzed a scenario under uncongested traffic conditions where CHVs could have greater freedom to choose between maintaining their longitudinal driving or changing lanes. To explore this scenario, we set the CAV penetration rate P_1 to 60% and the on-ramp flow ratio η_{ramp} to 20%. We considered five cases with different CHV compliance rates (0%, 25%, 50%, 75% and 100%). Table II presents the average flow and speed of the mainline and merging bottleneck under these different CHV compliance rate, as well as the corresponding percentage difference compared to the reference case with 0% compliance rate. The results show that the mainline and merging flows, as well as the mainline and merging speed, increase with the increase of the compliance rate, except for a slight decrease in merging flow and merging speed at 25% and 50% compliance rates, respectively. The performance improvements in terms of flow and speed are marginal when the compliance rate is less than 50%, while the benefits of the CUC model become more significant when the CHV compliance rate is larger than 75%.

C. Impact of CAV Penetration Rate on Mainline Traffic Efficiency

In this section, we discuss the impact of CAV penetration rate on mainline traffic efficiency in terms of average flow and average speed. As shown in Table II, a CHV compliance rate higher than 50% could lead to a good performance

TABLE III

AVERAGE MERGING AND MAINLINE FLOWS WITH DIFFERENT CAV PENETRATION RATE AND ON-RAMP FLOW RATIO (COMPLIANCE RATE IS 75%)

η_{ramp}	Flow (veh/h/lane)	P_1 (%)					
		0%	20%	40%	60%	80%	100%
10%	Merging flow	1066	1201	1332	1497	1609	1893
	Diff (%)	-	12.7	25.0	40.4	50.9	77.6
	Mainline flow	1106	1192	1332	1531	1683	1962
	Diff (%)	-	7.8	20.4	38.4	52.2	77.4
20%	Merging flow	1076	1228	1372	1469	1585	1886
	Diff (%)	-	14.1	27.5	36.5	47.3	75.3
	Mainline flow	1148	1309	1417	1616	1717	1997
	Diff (%)	-	14.0	23.4	40.8	49.6	74.0
30%	Merging flow	1076	1158	1264	1409	1655	1877
	Diff (%)	-	7.6	17.5	30.9	53.8	74.4
	Mainline flow	1144	1218	1423	1525	1752	1955
	Diff (%)	-	6.5	24.4	33.3	53.1	70.9
40%	Merging flow	979	1097	1137	1386	1645	1863
	Diff (%)	-	12.1	16.1	41.6	68.0	90.3
	Mainline flow	1031	1175	1254	1454	1657	1832
	Diff (%)	-	14.0	21.6	41.0	60.7	77.7

improvement. Therefore, in this experiment, the compliance rate is set to be 75%. Table III presents the average flow of the mainline and merging bottleneck under different CAV penetration rates and on-ramp flow ratios, as well as the corresponding percentage differences compared to the reference case of 0% CAVs. It is clear that the mainline and merging flows increase significantly with the increase of the CAV penetration rate under different on-ramp flow ratios (from 10% to 40%). For example, under the on-ramp flow ratio of 10%, the percentage increase of the mainline flow and that of the merging flow range from 12.7% to 77.6% and from 7.8% to 77.4%, respectively, when the penetration rate increases from 20% to 100%. Similar results can be found for the cases of 20% - 40% on-ramp flow ratio. Additionally, both the mainline and merging flows first increase, and then decrease as the on-ramp flow ratio increases for a given CAV penetration rate. This indicates that the traffic condition of the merging area changes from the uncongested to congested condition when η_{ramp} increases from 10% to 40%.

Table IV shows the average speeds of the mainline and merging bottleneck, percentage speed difference, and corresponding merging and mainline flow rate under different CAV penetration rates and on-ramp flow ratios. It is clear that the average speed increases with the increase of the penetration rate. For the CAV penetration rate smaller than 60%, the increase of CAV penetration rate has a marginal impact on the increases of the average mainline and merging speeds under different on-ramp flow ratios, whereas the impact is more significant for higher CAV penetration rates (>60%). The possible reason is that when the CAV penetration rate is low, only a limited number of vehicles in the mainline adopt the proposed CORMC model to improve their driving conditions, resulting in a marginal influence on the overall speed

TABLE IV

AVERAGE SPEEDS OF THE MAINLINE BOTTLENECK AND MAINLINE WITH DIFFERENT CAV PENETRATION RATE AND ON-RAMP FLOW RATIO (COMPLIANCE RATE IS 75%)

η_{ramp}	Speed and flow (veh/h/lanes)	P_1 (%)					
		0%	20%	40%	60%	80%	100%
10%	Merging speed	25.4	25.6	26.3	26.3	26.5	29.2
	Diff (%)	-	0.8	3.5	3.5	4.3	15.0
	Mainline speed	24.8	25.5	26.2	26.4	27.2	29.8
	Diff (%)	-	2.8	5.6	6.5	9.7	20.2
	Merging flow	1066	1201	1332	1497	1609	1893
	Mainline flow	1106	1192	1332	1531	1683	1962
20%	Merging speed	20.6	21.1	21.2	21.4	23.3	27.4
	Diff (%)	-	2.4	2.9	3.9	13.1	33.0
	Mainline speed	23.7	23.9	24.8	26.1	27.1	29.8
	Diff (%)	-	0.8	4.6	10.1	14.3	25.7
	Merging flow	1076	1228	1372	1469	1585	1886
	Mainline flow	1148	1309	1417	1616	1717	1997
30%	Merging speed	15.2	15.8	16.0	15.8	17.8	23.4
	Diff (%)	-	3.9	5.3	3.9	17.1	53.9
	Mainline speed	22.1	22.4	24.6	24.9	28.1	29.8
	Diff (%)	-	1.4	11.3	12.7	27.1	34.8
	Merging flow	1076	1158	1264	1409	1655	1877
	Mainline flow	1144	1218	1423	1525	1752	1955
40%	Merging speed	14.0	14.4	15.1	15.7	18.2	21.2
	Diff (%)	-	2.9	7.9	12.1	30.0	51.4
	Mainline speed	20.2	22.0	23.4	23.5	25.7	29.8
	Diff (%)	-	8.9	15.8	16.3	27.2	47.5
	Merging flow	979	1097	1137	1386	1645	1863
	Mainline flow	1031	1175	1254	1454	1657	1832

improvement. In addition, the improvement of speeds for both the merging bottleneck and mainline is more significant when the on-ramp flow is relatively high (e.g., $\eta_{ramp} = 40\%$). The percentage increase of the mainline speed and that of the merging speed range from 8.9% to 47.5% and from 2.9% to 51.4%, respectively, when the CAV penetration rate increases from 20% to 100%. This indicates that the proposed CORMC model works more effectively under the situation where the merging bottleneck becomes congested (e.g., when the on-ramp flow ratios are 30% and 40%).

D. Performance Comparison Among Different Sub-Models

In this section, we discuss the performance of three control strategies:

Strategy 1: our proposed CORMC model with both the CUC model and the APS algorithm

Strategy 2: the CORMC model without the CUC model but with the APS algorithm

Strategy 3: the CORMC model without the APS algorithm but with the CUC model

Fig. 11 shows the average flow and speed at the merging bottleneck (Fig. 11 (a)(b)) and the mainline (Fig. 11 (c)(d)) for

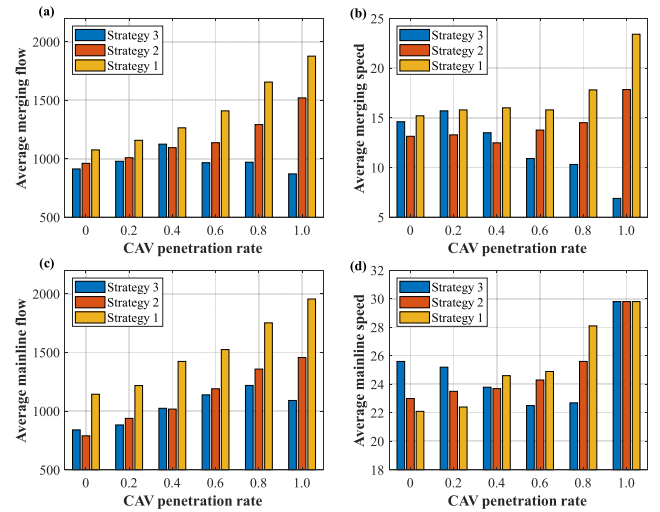


Fig. 11. Comparison of average flow and speed with different strategies when $\eta_{ramp} = 30\%$ and $C = 75\%$: a) average merging flow; b) average merging speed; c) average mainline flow; d) average mainline speed.

the three strategies. The on-ramp flow ratio η_{ramp} and CHV compliance rate (C) are set to be 30% and 75%, respectively. It can be clearly observed that strategy 1 outperforms the other two strategies in terms of both average flow (throughput) and average speed. Additionally, strategy 2 significantly outperforms strategy 3 across all CAV penetration rates. The average merging speed under strategy 3 drops sharply when the CAV penetration rate increases as can be seen from Fig. 11 (b). This indicates that, in the CORMC model, the APS algorithm plays a more critical role than the CUC model. Assigning a specific CV to the MV can significantly improve the merging speed. Comparing the throughput of strategy 2 and strategy 3 in Fig. 11 (a), we observe a marginal difference at low CAV penetration rates ($P_1 < 0.4$), but strategy 2 performs much better than strategy 3 at higher CAV penetration rates. This is because more HVs with larger headway gaps exist at low CAV penetration rates, so that the advantages of the APS algorithm cannot be fully exploited. In Fig. 11 (d), we observe that the average mainline speed of strategy 3 is the highest among the three strategies when the CAV penetration rate is very low (e.g., $< 20\%$). However, strategies 1 and 2 outperform strategy 3 when the CAV penetration rate is larger than 40%. When the CAV penetration rate is 100%, all vehicles are controllable, and the average mainline speed is the same for all three strategies. One possible reason for the observed results is that, at low CAV penetration rate, there are more CHVs with larger headway gaps (the desired time gaps of HVs and CAVs are 2.0s and 1.2s in the simulation experiments). As a result, merging can be successfully completed without the APS algorithm, and the presence of the APS algorithm could actually reduce the speed of mainline vehicles.

E. Comparison With SUMO

In this section, we verify the performance of the CORMC model by comparing it with the SUMO simulation model. The lane changing behavior in the SUMO strategy is described by the LC2013 model, which is also known as a strategic lane

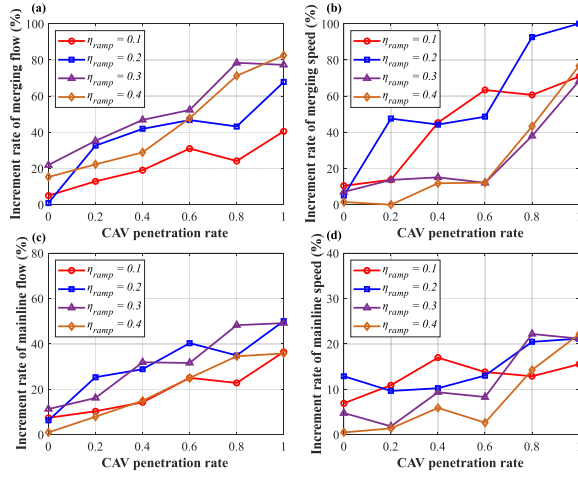


Fig. 12. The increment rates of average flow and speed under different CAV penetration rates compared to SUMO: a) merging flow; b) merging speed; c) mainline flow; d) mainline speed.

changing model [42]. This model can describe lane-changing behavior at lane drops or ramp bottlenecks. In the LC2013 model, a vehicle first evaluates the nearby lane as it approaches the area and determines the urgency of lane changing based on factors such as the remaining distance to the end of the area, speed, and traffic occupancy in the target lane. Next, the feasibility of lane changing is assessed based on the traffic conditions in the surrounding environment. The vehicle then adjusts its speed to safely execute the lane changing maneuver. It is worth noting that the LC2013 model does not include a mechanism for early cooperation with mainline vehicles before entering the on-ramp area.

In the scenario of SUMO simulation, the default driver behavior models, which are included in SUMO, are applied to all vehicles, while the scenario settings and traffic flow inputs are consistent with those in the proposed CORMC model. We define two indicators, the average speed increment rate (I_v) and the average flow increment rate (I_q), to measure the improvement achieved by the CORMC model, as given by:

$$I_v(A) = \frac{v_{CORMC}(A) - v_{SUMO}(A)}{v_{SUMO}(A)} \cdot 100\%$$

$$I_q(A) = \frac{q_{CORMC}(A) - q_{SUMO}(A)}{q_{SUMO}(A)} \cdot 100\% \quad (64)$$

where, $v_{CORMC}(A)$ and $q_{CORMC}(A)$ represent Eddie's generalized average speed and flow of the CORMC model in the observed time-space region A , while $v_{SUMO}(A)$ and $q_{SUMO}(A)$ represent those of the SUMO model.

Fig. 12 illustrates the increment rates of average flows and speeds of the mainline and merging bottleneck under different CAV penetration rates and on-ramp flow ratios compared to the SUMO strategy. As the CAV penetration rate increases, the flow and speed increment rates continue to rise under different on-ramp flow ratios, ranging from 10% to 40%. For instance, when $P_1 = 100\%$ and $\eta_{ramp} = 40\%$, the CORMC model can potentially increase merging throughput and speed by over 80%. These results demonstrate that the proposed CORMC model outperforms SUMO in improving the traffic

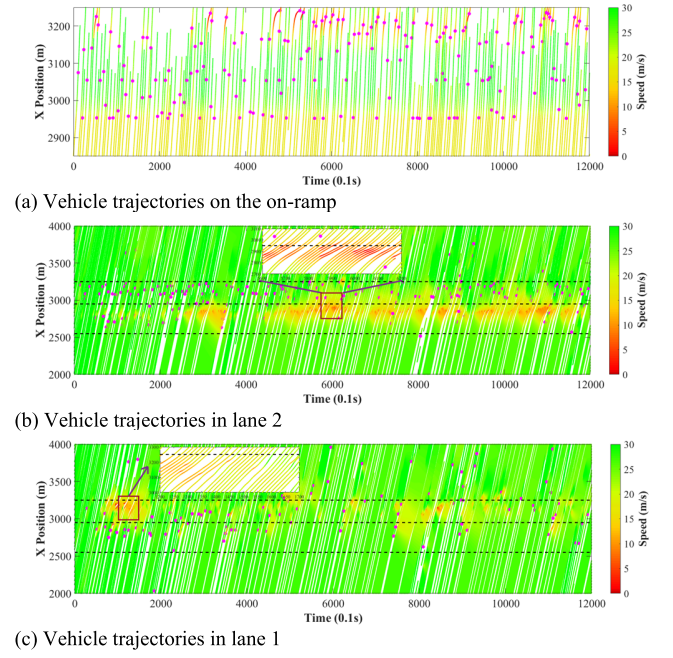


Fig. 13. Trajectories of all vehicles with $P_1 = 80\%$, $\eta_{ramp} = 20\%$ and $C = 75\%$ (CORMC).

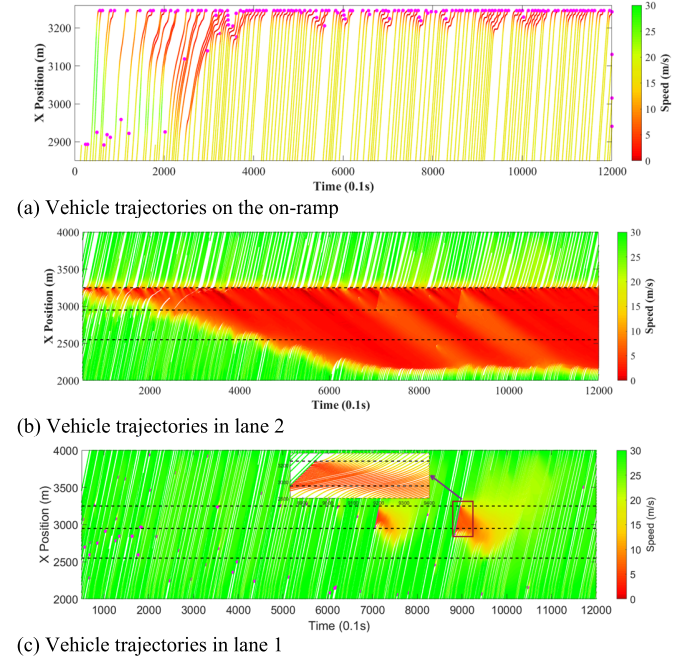


Fig. 14. Trajectories of all vehicles with $P_1 = 80\%$, $\eta_{ramp} = 20\%$ (SUMO).

efficiency in the on-ramp merging bottleneck area. Moreover, Fig. 12 (a)(c) shows that higher on-ramp ratios ($>20\%$) and CAV penetration rates ($>60\%$) lead to greater improvements in merging and mainline flows. This finding indicates that the CORMC model performs more effectively in situations where the merging bottleneck becomes congested.

Figs. 13 and 14 illustrate the vehicle trajectories of the CORMC and SUMO strategies, respectively, for scenario $P_1 = 80\%$ and $\eta_{ramp} = 20\%$. In Fig. 13, the pink dot represents the starting point of lane-changing on the corresponding lane,

while the two areas formed by three black dashed lines denote the merging and cooperative zones from top to bottom, respectively, as shown in Fig. 13(b)(c). Notably, no collisions occur during the entire process, indicating that the proposed CORMC model ensures safety. Fig. 13(a) reveals that the merging start positions of all MVs are distributed within the merging zone in the area between 2950m and 3250m, and the average merging speed is nearly 25m/s. Furthermore, all MVs can merge into lane 2 upstream of the downstream boundary of the on-ramp, validating the effectiveness of the proposed boundary-collision-avoidance algorithm. In contrast, for the SUMO simulation scenario, almost all MVs merge into lane 2 at the end of on-ramp (3250m) as shown in Fig. 14 (a), and the average merging speed is less than 18m/s. This demonstrates the superior efficiency of the proposed CMC model.

Figs. 13(b) and 14(b) demonstrate the impact of MVs on traffic dynamics of lane 2, such as speed reduction. This impact is due to the cooperative maneuvers of mainline vehicles, which decelerate to create gaps for the MV to merge into the mainline. In Fig. 14 (b), as on-ramp vehicles merge into lane 2, vehicles on lane 2 continue to decelerate, with some even stopping at the end of the merging zone. The deceleration wave then propagates upstream, gradually expanding. However, when the CORMC model is adopted, the impact is mainly in the cooperative zone upstream of the on-ramp. The deceleration wave caused by the merging does not propagate upstream and dissipates in a short period, as shown in Fig. 13 (b). This highlights the effectiveness of the proposed CORMC model in mitigating the propagation of congestion waves.

Furthermore, Fig. 14(c) depicts that when using the SUMO simulation, only a few vehicles changed lanes to lane 1 in the cooperative and merging zones, likely due to the absence of a proper lane change guidance model. While adopting the proposed CORMC model, there was a significant increase in the number of vehicles in lane 2 changing to lane 1 to create larger merging gaps for MVs, as shown in Fig. 13(c). Despite the lower speed on lane 1 in the CORMC model compared to the SUMO strategy, the average speed of both lanes was approximately twice that of the SUMO simulation, as shown in Fig. 12(b). This improvement in speed is due to the more balanced traffic flow between lanes in the cooperative and merging zones when using the CUC model compared to the SUMO simulation.

VII. CONCLUSION AND FUTURE WORK

In this paper, we present a CORMC model for multilane mixed traffic on freeways, which considers both longitudinal and lateral movements, where a hierarchical framework consisting of the APS algorithm, CUC model and CMC model is developed. The multi-lane freeway section is divided into three zones: mainline zone, cooperative zone and merging zone. In the mainline zone, longitudinal and lane-changing models are used to describe vehicles' movements. In the cooperative zone, the APS algorithm is developed to calculate anticipatory positions at which MVs merge from the on-ramp lane to the adjacent main lane (lane 2) and assign CVs in

lane 2 for MVs. Then, the CUC model is formulated to determine the best cooperative maneuver choices for CVs for facilitating the creation of merging gaps. In the merging zone, the CMC model coordinates the movements of vehicles on the on-ramp and adjacent lane 2 for safe and efficient merging. A boundary-collision-avoidance algorithm is proposed to guarantee collision-free between MVs and the downstream boundary of the on-ramp.

To verify the effectiveness of the proposed models and algorithms, we develop a micro-simulation platform using MATLAB. A vehicle generation model is proposed to simulate vehicle arrivals, and the uncertainty and heterogeneity of traffic flow is explicitly considered. Based on numerical experiments, we make six main observations. First, the performance benefits brought by the CUC model are marginal when the CHV compliance rate is low ($<50\%$), while the performance improvement is significant with higher compliance rates ($>50\%$). Second, the impact of the low CAV penetration rates ($<60\%$) on the performance improvement of the mixed traffic system is marginal under different on-ramp flow ratios, whereas the impact is more significant for higher CAV penetration rates ($>60\%$). Third, the improvement of speed for the merging bottleneck and mainline is more significant in congested conditions when the on-ramp flow is relatively high (e.g., $\eta_{ramp} = 40\%$). Fourth, the proposed CORMC model (Strategy 1) outperforms the CORMC without the CUC model (Strategy 2) and the CORMC without the APS algorithm (Strategy 3) in terms of throughput and average speed. The comparison results indicate that the APS algorithm plays a more important role in the CORMC model than the CUC model. The APS algorithm assigns a specific CV to the MV, which could greatly improve the merging speed. Fifth, there are no collisions during the entire process, indicating that the safety of vehicles using our proposed CORMC model can be assured. Finally, the proposed CORMC model can effectively mitigate the propagation of congestion waves.

In future, we aim to extend the proposed CORMC model to freeways with more than two mainline lanes, taking into account the imbalance of flow and the difference in equilibrium speed on different mainline lanes. Additionally, with the proposed hierarchical control framework, we plan to investigate control strategies in the multi-lane freeway off-ramp bottleneck area.

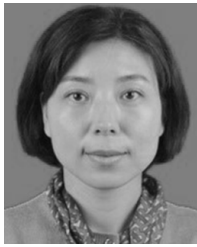
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